

Soccer Robot

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This soccer robot can move forward, reverse, forward-left, forward-right, reverse-left and reverse-right with the help of an Android phone. The speed of its movement is controlled by the angle

Circuit and working

Fig. 3 shows the circuit of the soccer robot. The circuit is built around Arduino UNO board (BOARD1), servo motor (connected at CON5), Bluetooth module JY MCU BT, motor driver L293D (IC1) and two DC motors (M1 and M2).

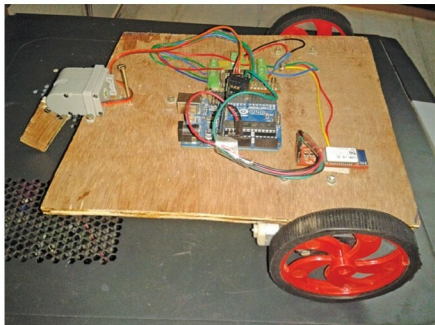


Fig. 1: Author's prototype

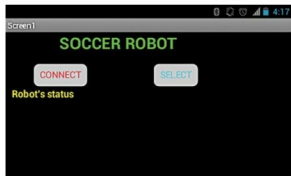


Fig. 2: Android app interface

of rotation of the phone. The robot also kicks a ball when the phone is shaken. Fig. 1 shows the author's prototype of the robot and Fig. 2 shows the app running on an Android phone to control the robot.

The circuit uses two 9V batteries; one for the Arduino board (BOARD1) and the other for DC motors. The 5V regulated supply for rest of the circuit is provided by Arduino board itself. LED on BOARD1 indicates the presence of power supply.

All new smartphones contain an accelerometer sensor. A phone with an accelerometer can detect the angle it is being held at. It can also detect rotation and motion gestures such as swinging, shaking and flicking. The accelerometer can detect whether the phone is

upright or sideways, and automatically rotate the graphics on the screen accordingly. Its another common use is in controlling games and other applications (such as music player) by moving or shaking the phone. When the phone is rotated in different directions, x,y,z values of accelerometer sensor change. The Android app for this project sends these x,y,z values to the robot via Bluetooth.

Bluetooth module. The Bluetooth module used in this project is JY MCU BT, which can be connected to any device via built-in UART interface. It can communicate with other Bluetooth-enabled devices such as mobile phones, handheld computers and laptops. The module runs on a 3.6V to 6V supply. Fig. 4 shows a picture of this Bluetooth module.

Servo motor. A servo motor is used for kicking the ball in this project. Servos are extremely useful in robotics. The motors are small but extremely powerful for their size. A servo motor mainly comprises a DC motor, gear system, position sensor (which is generally a potentiometer) and control electronics as shown in Fig. 5.

The DC motor is connected with a gear mechanism which provides feedback to a position sensor. The potentiometer allows the control circuitry to monitor the current angle of the servo motor. If the shaft is at correct angle, the motor shuts off. If the circuit finds that the angle is incorrect, it keeps the motor on until the angle is corrected. A normal servo is used to control an angular motion up to 180 degrees. Servos are incapable of turning further due to a mechanical stop built on the main output gear.

The control wire is used to communicate the angle. The angle is determined by the duration of a pulse that is applied to the control wire. This is called pulse-coded modulation. The servo expects to see a pulse

